

Abhinav Kannan

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EDUCATION

- **Indian Institute of Science Education and Research (IISER), Pune** Aug 2021 – Present
BS-MS in Biology CGPA: 8.1
- **Chinmaya Vidyalaya Higher Secondary School, Chennai** 2021
Central Board of Secondary Education (Grade 12) Percentage: 97.2%

RESEARCH EXPERIENCE

- **Eco-Evolutionary Dynamics of Cancer Stem Cells in Spatially Explicit Environments** June 2025 – Present
Master's Thesis Student with Prof. Dominik Wodarz UC San Diego
 - Developing models of feedback and feedforward regulation within spatially explicit cell lineages to investigate how evolutionary processes affect the maintenance and adaptation of cancer stem cells (CSCs) in spatially regulated systems
 - Analyzing how interactions between CSCs and the extracellular matrix influence the evolution of CSC phenotypes under spatial and regulatory constraints
- **Investigating the role of the V-domain in regulating Drp1's binding to its adaptor** Aug 2024 – Dec 2024
Undergraduate Researcher with Prof. Thomas Pucadyil ([Report](#)) IISER Pune
 - Characterized the regulatory role of a specific protein-binding domain in Drp1's interaction with mitochondrial adaptor proteins using biochemical approaches
 - Gained expertise in recombinant protein expression in *E. coli*, affinity chromatography-based purification, molecular cloning (PCR, sequencing) and protein-protein interaction assays (dot-blot)
- **Systems-level analysis of cellular plasticity in ER+ breast cancer** May 2023 – Aug 2024
Undergraduate Researcher with Prof. Mohit Kumar Jolly ([Manuscript in preparation](#)) IISc Bangalore
 - Developed and analyzed a minimalistic regulatory network model of cancer cell plasticity using both discrete (Boolean) and continuous (ODE-based) simulations
 - Investigated the high-dimensional phenotypic space to identify coupled transitions between epithelial-mesenchymal states, metabolic reprogramming, therapy resistance, and immune evasion

SKILLS

- **Computational Modeling:** Agent-based modeling, stochastic cellular automata, ODE/PDE systems, Gillespie simulations, network dynamics
- **Programming:** Julia, Python, Bash, LaTeX
- **Experimental Methods:** Protein purification, cloning, PCR, Western blotting
- **Languages:** English (fluent), Tamil (native), Hindi (proficient), Sanskrit (basic)

RELEVANT COURSEWORK

- **Biology:** Animal Behavior, Bioinformatics, Cell Biology, Computational & Functional Genomics, Developmental Biology, Ecology, Evolution, Genetics, Immunology, Mathematical & Computational Biology, Molecular Biology, Neurobiology, Physiology, Systems Biology
- **Mathematics:** Algorithms, Applied Mathematical Methods, Bayesian Theory, Data Analysis, Graph Theory, Probability Theory, Statistical Learning, Signal Processing, Multivariate Calculus, Numerical Linear Algebra
- **Physics:** Mathematical Methods for Physics, Nonlinear Dynamics, Thermodynamics, Statistical Mechanics

WORKSHOPS AND PROFESSIONAL TRAINING

- **Workshop – Unifying Theories in High-Dimensional Biophysics** July 2025
Selected Participant International Centre for Theoretical Sciences, Bangalore
– Engaged with interdisciplinary frameworks for interpreting high-dimensional biological data across development, neuroscience, immunology, and evolution and explored connections between low-dimensional modeling, machine learning, and emergent biological simplicity through a series of pedagogical lectures and research talks
- **Workshop – Decisions, Games and Evolution** March 2025
Selected Participant International Centre for Theoretical Sciences, Bangalore
– Studied the evolution of cooperation, decision-making, and collective behavior using evolutionary game theory, network dynamics, and learning strategies through a series of pedagogical lectures and research talks
- **Workshop – Spins, Games and Networks** Dec 2024
Selected Participant Institute of Mathematical Sciences, Chennai
– Completed a two-week long group project titled "Simulating Social Dynamics in Party-like Scenarios" using network-based modeling ([Presentation slides](#))
– Studied game theory, evolutionary dynamics, complex networks, time-series analysis, statistical physics, and deep neural networks through a series of pedagogical lectures and research talks
- **Workshop – Flags, Landscapes and Signaling** May 2024
Selected Participant Institute of Mathematical Sciences, Chennai
– Studied developmental pattern formation, their breakdown leading to cancer, and how these mechanisms might change over the course of evolution through a series of pedagogical lectures and research talks

SCHOLARSHIPS AND HONORS

- **INSPIRE-SHE Scholar** ($\approx 4,500$ USD) – Awarded by the Department of Science and Technology, Government of India, for academic excellence in science 2021–Present
- **IISER Aptitude Test (IAT)** – All India Rank 200 / 20,000 2021

REFERENCES

Prof. Dominik Wodarz (UC San Diego)
Email: dwodarz@ucsd.edu

Prof. Mohit Kumar Jolly (IISc, Bangalore)
Email: mkjolly@iisc.ac.in